

Part 1. Metadata

- Version: design (25/3/2024)
- Version: Implementation (17/05/2024)
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- Students: 
- Group number: 
- Dataset: **SunCharge**

Part 2. Project description

1. A short description of what the data/project is about (max 3 sentences)

The project centers on SunCharge, a company specializing in eco-friendly cars and home batteries, with a primary aim of optimizing the supply chain for smooth production processes and timely product delivery while meeting customer demands effectively. Using data visualization design, the goal is to transform complex supply chain data into intuitive graphics. The dataset covers various supply chain aspects, aiming to develop a monitoring system that detects issues and informs decision-making for optimizing logistics operations.

2. Description of some selected features:

- **Material:** Identifies the specific materials used in production. It is connected to other features such as Component and MaterialType, providing information on the composition and classification of materials within the build of materials (BOM).
- **Quantity-related features:** Quantitative features are vital for optimizing supply chain operations. *GrossInventoryQuantity* offers an overview of total inventory, while *OnShelfInventoryQuantity* ensures sufficient stock availability for immediate use. *OnTransitQuantity* tracks goods in transit, aiding logistics coordination. These features, linked to the Inventory datasheet, enable SunCharge to allocate resources efficiently, minimize stockouts, and enhance overall performance.
- **Date-related features:** Date-related features are crucial for efficient supply chain operations. *PurchaseOrderCreationDate* and *SalesOrderCreationDate*, for instance, are linked to the Purchases and Sales datasheets respectively provide insights into procurement and sales timing, facilitating order monitoring and processing efficiency. *PlannedGoodsReceiptDate* and *ActualGoodsReceiptDate* are connected to the Inventory datasheet, and track inbound shipments, aiding warehouse management. Additionally, *RequestDeliveryDate*, found in the Forecast datasheet, prioritizes order fulfillment, while *PlannedArrivalDateYard*, linked to the Purchases datasheet, coordinates inbound logistics and optimizes inventory. Together, these features streamline SunCharge's supply chain processes, enhancing operational performance and delivering value to customers.
- **OrderType:** By classifying orders into "Express" and "Regular" types, SunCharge can effectively prioritize and manage its order fulfillment processes. Express orders receive heightened attention to meet urgent customer demands, while regular orders are processed according to standard procedures, ensuring efficient utilization of resources and timely delivery across the supply chain.
- **Keys:** *PlantKey* uniquely identifies production plants, linking data across various sheets like CustomerPlantRelation, Inventory, MaterialPlantRelation, Purchases, and Sales, tracking plant-specific details such as inventory levels and procurement activities. *VendorKey* uniquely identifies vendors in the Vendors sheet and provides distinction whether a vendor is an external or internal vendor, connecting to MaterialPlantRelation and Purchases for monitoring vendor-specific data like purchase history and delivery performance. *MaterialKey* uniquely identifies materials in the Materials sheet, linking to BOM, Forecast, Inventory, MaterialPlantRelation, Purchases, and Sales, facilitating tracking of material-specific data such as usage patterns and inventory levels. These keys streamline procurement, inventory management, and sales processes, enhancing overall operational efficiency and customer satisfaction throughout the supply chain.

3. 2 or 3 questions that guide you in the rest of the project.

1. How can we leverage data visualization techniques to identify bottlenecks and inefficiencies within SunCharge's supply chain, ultimately improving operational performance?
2. What strategies can be implemented to enhance inventory management practices and minimize stockouts, ensuring seamless production processes and timely product deliveries?
3. How can we optimize logistics operations to streamline the movement of goods and reduce lead times, thereby enhancing overall supply chain agility and responsiveness?

Part 3. Visual design

1. A description of the process, how you created the final designs (which approach, how did you collaborate, ...)

Our visual design process for SunCharge's supply chain optimization project was characterized by collaboration and iteration, leveraging various tools and mediums to explore and refine our ideas. We began with the diverge phase, where we employed divergent thinking to explore a wide range of design possibilities independently. Leveraging hand-drawn sketches for initial ideation, we applied techniques such as reworking existing visuals, combining dissimilar ideas, and organizing our exploration using matrix-based approaches.

As we progressed into the emerge phase, we refined our initial sketches collaboratively, employing the SCAMPER technique to further develop and enhance our design ideas. Additionally, we utilized video call meetings for real-time discussions, enabling immediate feedback and fostering collaboration between team members (JA and SH).

Furthermore, Miro served as a crucial platform for virtual collaboration, allowing us to explore, share, and rework sketches in real-time. Google Drive and Github were also integral to our process, serving as platforms for file-sharing. These tools facilitated seamless access to design assets, ensuring that we could easily collaborate on and access the latest versions of our designs. Throughout this phase, we continually explored different design permutations, combining and refining our concepts to identify the most promising solutions.

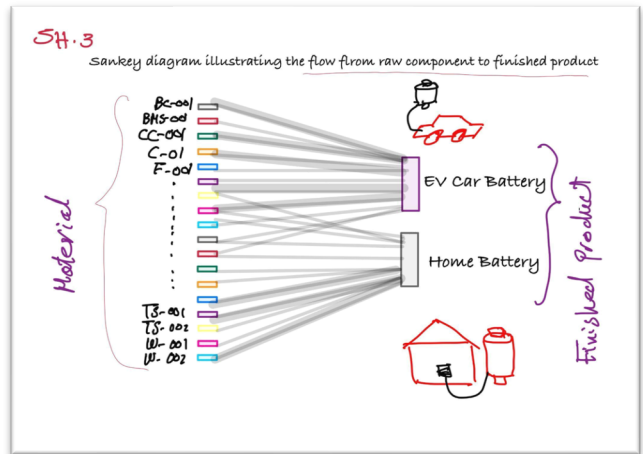
In the converge phase, we integrated our designs, converging on a set of final visualizations that effectively communicated insights from the supply chain dataset. This integration involved selecting the most effective design elements from our exploration and integrating them into cohesive and impactful visualizations. By leveraging a combination of divergent thinking, SCAMPER, matrix-based approaches, real-time discussions via video calls, and effective file-sharing through Google Drive, Github, and collaborative sketching on Miro, the team members navigated through the iterative design process, ultimately delivering solutions that addressed SunCharge's supply chain challenges.

2. Five to 10 sketches that you think best illustrate the path you took from diverge to emerge and converge. Explain their encoding and why you chose them.

Sketch 1 (SH_3): Sankey Diagram

- Nodes: Rectangles with labels denote raw materials and finished products, with each finished product represented by an icon (car for EV Car Battery, house for Home Battery).
- Links: Flowing lines indicate material flow from raw components to finished products, with thickness reflecting material quantity.

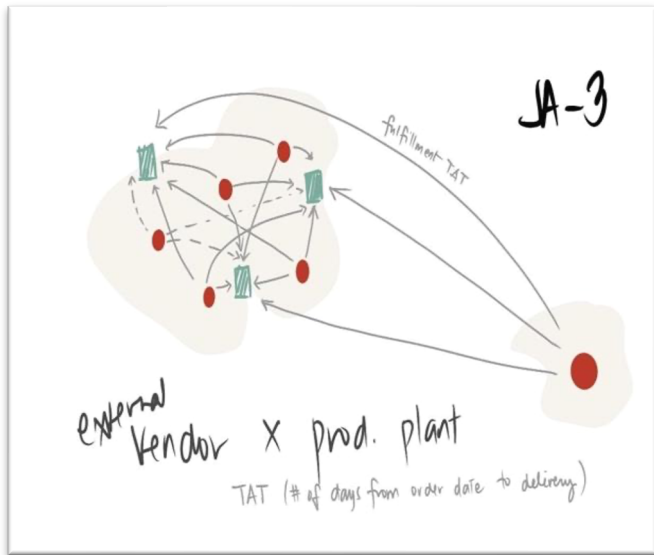
This Sankey Diagram illustrates materials flow in EV Car Battery and Home Battery manufacturing, highlighting resource dependencies and environmental impacts. Featuring 33 materials, including 9 common to both products, and visual icons for easier comprehension, it aids in addressing sustainability, resource efficiency, and supply chain management queries



Sketch 2 (JA_3): Vendor-to-Production Relationship Map

- Nodes: Red circles represent external vendors supplying materials, while green circles denote production plants where finished products are made.
- Lines: Gray lines with one-directional arrows indicate material flow from vendors to production plants, accompanied by Total Inbound Time annotations showcasing the average production duration for materials supplied by each vendor.
- Visual Insights: The map's shape and spatial distribution offer a comprehensive overview of the purchasing process, providing valuable insights into geographic distribution and distances between vendors and production plants.

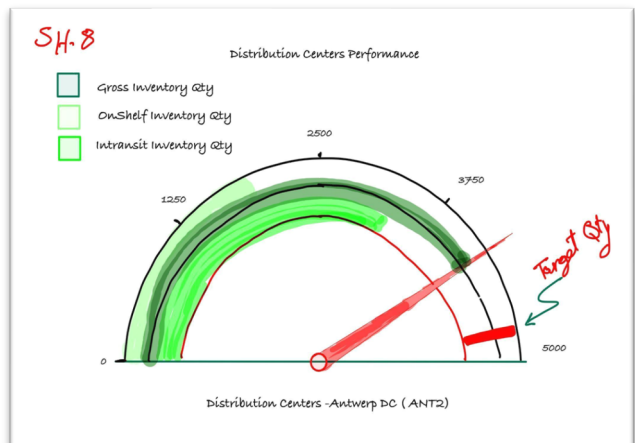
This visualization provides a holistic understanding of the vendor-production plant relationship, offering insights into order fulfillment times based on distance and location. By illustrating material flow and production durations, it aids in comprehending supply chain dynamics and addressing queries regarding efficiency, logistics optimization, and resource allocation.



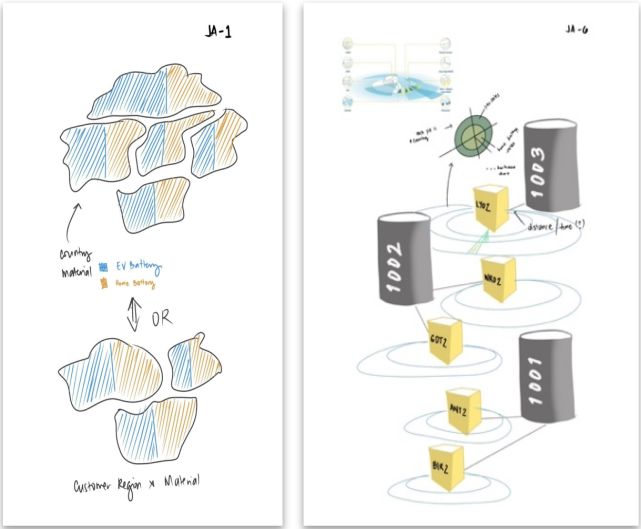
Sketch 3 (SH_8): Gauge Visualization with Legend

- Gauge: Circular dial with a needle indicating performance level, distinguished by color shades representing gross inventory quantity, on-shelf inventory quantity, and in-transit inventory quantity.
- Colors: Used to differentiate between metrics, with distinct shades for each inventory category.
- Target Mark: Represents the target performance level or benchmark for easy reference.
- Enhanced Interpretation: The inclusion of a legend improves visualization and readability by providing a key for interpreting gauge colors.

Insightful Visualization: This reworked Gauge Visualization with Legend offers a comprehensive representation of distribution center performance. Users can quickly grasp performance across various inventory metrics, aided by interactive features for detailed analysis and trend tracking over time.



Sketch 4(JA_1) & Sketch 5 (JA_6): Sales by Country and the Relationship between Production and Distribution



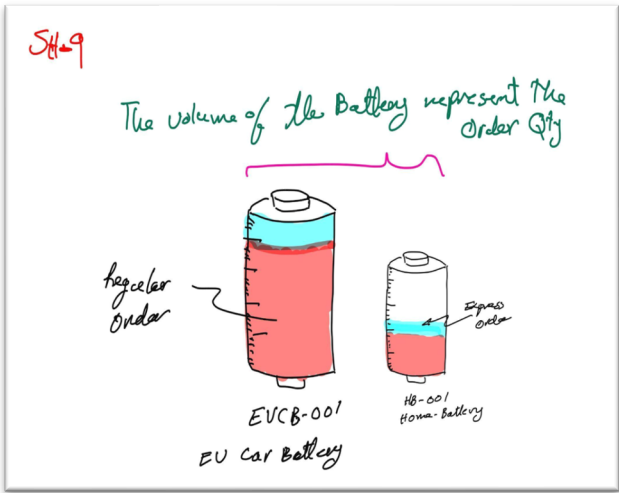
- Sketch 4(JA_1) presents the sales proportion for each finished product (EV Battery or Home Battery) per country, offering insights into product preferences across different regions.
- Sketch 5(JA_6) offers a 3D visualization depicting the relationship between production plants and the distribution centers they serve. Orbiting lines represent the distance of distribution centers from each country they cover, with attempts made to illustrate sales volume for each product per country. Note that 3D visualizations may be revised to a 2D format for clarity.

These visualizations facilitate forecasting of future sales by revealing product demand trends and identifying geographical factors influencing sales patterns. By discerning marketable countries, adjustments to target inventory levels can be made at distribution centers accordingly. Moreover, for countries with underperforming products, targeted investigations can be conducted to uncover the reasons behind the lackluster sales performance.

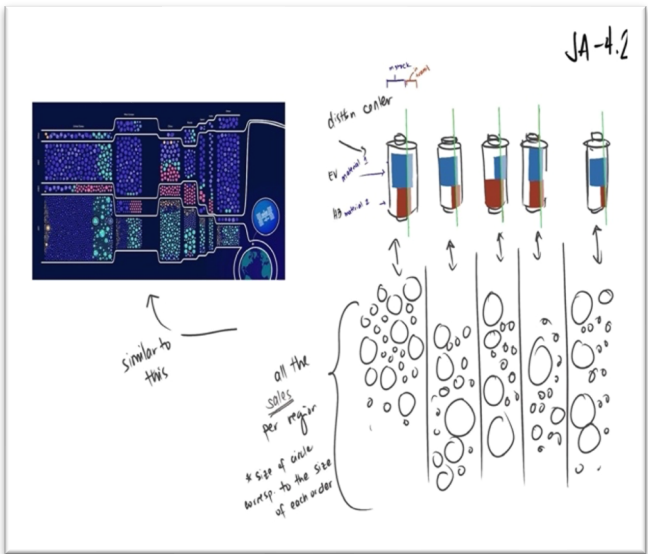
Sketch 6 (SH_9): Battery Visualization

- Batteries: Depicted as drawings of batteries, with one representing the home battery and the other the EV Car Battery.
- Colors: Two distinct colors indicate express and regular orders for each battery, with the color portion reflecting the volume of each order type.
- Volume of Batteries: The size of each battery corresponds to the total order quantity, with larger batteries indicating higher order volumes.

This redesigned Battery Visualization offers a clear representation of order distribution, distinguishing between express and regular orders through color codes. The size and color portion of each battery accurately depicts order volumes and priorities, aiding in understanding order patterns. Additionally, interactive features enable detailed exploration of order information, facilitating informed decision-making regarding order management and fulfillment strategies.



Sketch 7 (JA_4): Distribution Center Inventory and Sales

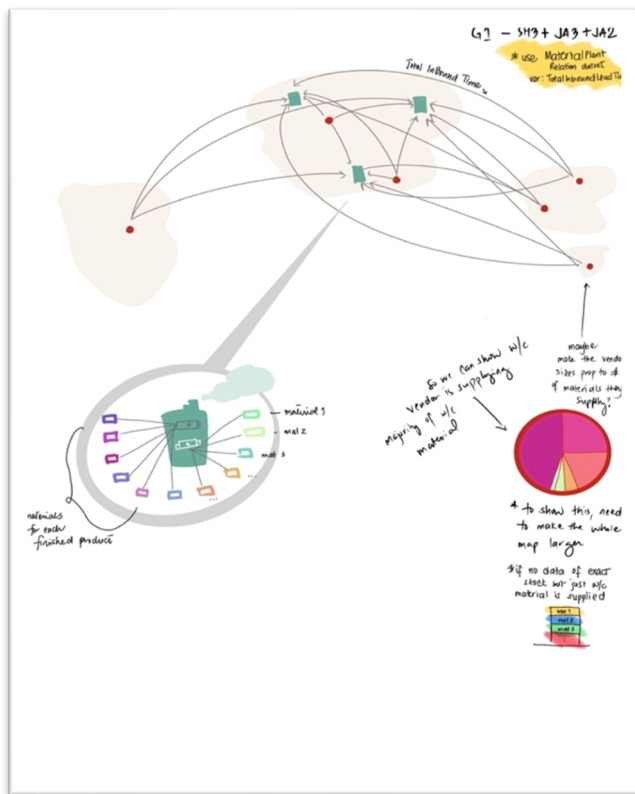


- The 5 battery shapes represent distribution centers (DCs), with vertical lines indicating the split between InStock and InTransit Inventory, and horizontal lines representing the split for each material held by the DC.
- Initially designed to just have lines splitting EV vs HB products (sketch JA_4) but then decided to put a bar chart inside the battery instead for more readability of the design (JA_4.2)
- Arrows illustrate the relationship between the DCs and the sales plot at the bottom.
- Below, the 5 splits are filled with circles of varying sizes, each representing a sales order. The size of the circle indicates the sales order quantity.

This visualization helps to identify which DCs have the largest stock of each product and to analyze the sales performance of products at each DC. It also aids in determining whether products at each DC are selling in large volumes or small volumes, facilitating strategic decisions such as stock redistribution to fulfill urgent orders.

This sketch draws inspiration from a space-related graphic about satellites (seen in the reference picture to the right).

3. Which 2-3 reworked sketches would you want to have implemented if you had an experienced programmer at your disposal?



Converge 1: Vendor-to-Production Relationship Map and Modified Sankey Diagram

- Nodes depicted as red circles representing external vendors supplying materials, and green rectangles symbolizing production plants where finished products are manufactured.
- Gray lines with one-directional arrows indicating material flow from vendors to production plants, accompanied by Total Inbound Time annotations showcasing the average production duration for materials supplied by each vendor.
- Visual insights provided by the map's shape and spatial distribution, offering a comprehensive overview of the purchasing process and insights into geographic distribution and distances between vendors and production plants.

The zoomed bubble from sketch SH_3 supplements the main plot by illustrating the materials combined to achieve the finished product:

- Nodes represented by rectangles with labels for raw materials and finished products, with each finished product depicted by an icon (car for EV Car Battery, house for Home Battery).
- Flowing lines indicating material flow from raw components to finished products, with thickness reflecting material quantity.

This combined visualization allows viewers to observe the entire process flow, from raw materials to finished products, incorporating location and time aspects, to provide insights regarding order fulfillment. It enhances understanding of supply chain dynamics, efficiency, logistics optimization, and resource allocation. Furthermore, the Sankey diagram illustrating the raw materials comprising the finished product addresses sustainability, resource efficiency, and supply chain management queries, rendering the final sketch a comprehensive tool for analyzing and optimizing production processes.

Converge 2: Production to Distribution Overall Performance

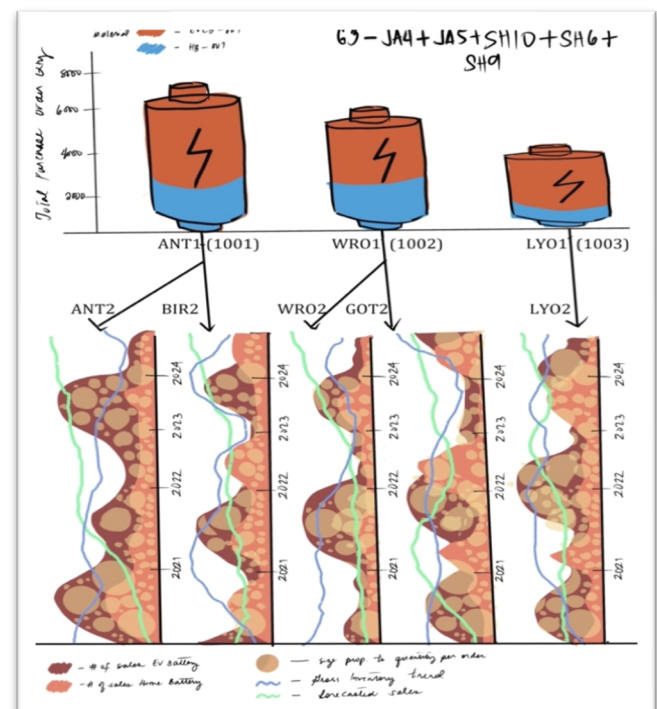
Converge 2 integrates elements from sketches JA_4, JA_5, SH_6, SH_9, and SH_10 to provide a comprehensive visualization of production plants, distribution centers (DCs), sales orders, and inventory trends.

The top part of the visualization is derived from JA_4, where batteries now represent production plants split by color signify the total purchase quantity order for each product. Linked downward arrows indicate the relationship between production plants and the 5 DCs.

The bottom section of the visualization features plots representing each DC, providing detailed insights:

- Solid maroon histograms depict the distribution of EV Battery sales orders, while superimposed coral-colored histograms represent Home Battery sales orders.
- Translucent circles represent sales orders, varying in size based on order quantity.
- Blue lines show the trend of Gross Inventory for all products at each DC over time.
- Neon green lines indicate the trend of forecasted sales over time.

These visualizations are plotted over time, allowing for the identification of seasonal patterns in orders and enabling comparison of performance between different DCs. This facilitates the identification of notable periods, such as months with high or low sales, and enables analysis of geographic impacts on sales and inventory status.



Part 4. Implementation

Introduction

In this section, we explore two custom visualizations developed during the design phase aimed at addressing pivotal questions for our project. These visualizations, derived from initial sketches in the Design Part, have undergone multiple rounds of feedback and iterative improvements to optimize their impact. Our primary objective was to create intuitive and interactive tools for efficiently monitoring and analyzing supply chain performance at SunCharge. Specifically, we focused on inventory management and distribution center performance (Implementation 1), along with plant shipment performance to distribution centers (Implementation 2). Each visualization is meticulously detailed, covering aspects such as intended design, actual implementation, visual encoding, interactions, and how they adhere to or diverge from the original sketches.

Tools and Software Used:

The visualizations were developed using the Svelte framework in conjunction with D3.js, utilizing Svelte REPL and SvelteKit in Visual Studio Code for creating interactive and dynamic components.

Implementation 1: Distribution Centers Performance Monitoring Tool

Description of the visual

The Distribution Centers Performance Monitoring Tool integrates two major components to provide a comprehensive view of the supply chain performance at SunCharge: the Battery Distribution Visualization and the Gauge Visualization for Inventory Performance. The Battery Distribution Visualization uses battery shapes to represent different distribution centers (DCs), with each battery divided into sections that depict the forecast quantities of EV Car Batteries and Home Batteries. These sections are filled based on the percentage of the forecast quantity, offering a clear and intuitive understanding of Sales predictions. The visualization dynamically updates based on the selected year and quantity threshold from drop-down menus, enabling users to view historical data and future forecasts. An alarm system at the battery positive terminal alerts users to potential issues by changing the flashing color: green indicates both inventory and forecasted quantities are above the threshold, orange indicates one falls below the target, and red when both fall below the threshold. Interactive features include hover effects displaying detailed information and a responsive.

The Gauge Visualization for Inventory Performance features gauges for each material type, representing different inventory metrics such as Gross Inventory Quantity, On-Shelf Inventory Quantity, In-Transit Quantity, and Order Quantity. The outer gauge represents EV Car Battery quantities, while the inner gauge represents Home Battery quantities. These gauges are color-coded, with needles and arcs indicating performance levels. They update dynamically based on the selected year, allowing users to compare current inventory and sales quantities with forecasted data depicted in the Battery Distribution Visualization. This comparative analysis helps evaluate the accuracy of sales and forecasts and identify discrepancies between expected and actual performance. The legend provides a key for interpreting the colors and metrics used, enhancing the visualization's clarity and usability.

Intended Design

The Distribution Centers Performance Monitoring Tool consists of two major components: the Battery Distribution Visualization and the Gauge Visualization for Inventory Performance. The intended design was inspired by multiple sketches from the Design Part, aiming to provide a comprehensive and interactive representation of supply chain data.

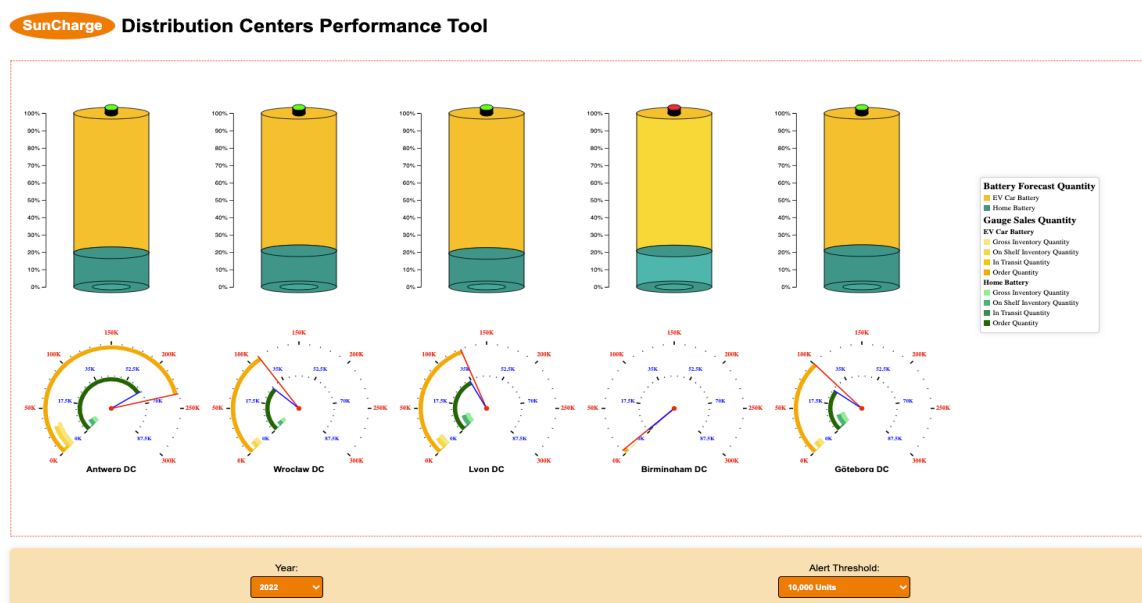
- **Sketch Reference:**
 1. **Sketch 6 (SH_9): Battery Visualization:**
 - Depicts batteries representing home and EV car batteries with distinct colors for express and regular orders.
 - The size of each battery reflects the total order quantity.
 2. **Converge 2: Production to Distribution Overall Performance**
 - Integrates elements from various sketches to visualize production plants, distribution centers (DCs), sales orders, and inventory trends.
 3. **Sketch 3 (SH_8): Gauge Visualization with Legend**
 - A circular dial with a needle indicating performance levels, differentiated by color shades representing various inventory metrics.

Actual Design

The implemented visualizations closely follow the intended design, incorporating dynamic and interactive features to enhance usability and insight generation. The tool was developed using the Svelte framework in conjunction with D3.js, enabling the creation of responsive and engaging visual components.

- **Battery Distribution Visualization:**
- **Visual Encoding:**
 1. **Battery Shapes:** Each distribution center is represented by a vertical cylinder (battery), with sections colored to indicate the forecast quantity of EV Car Battery (yellow) and Home Battery (teal).
 2. **Y-Axis:** Indicates the quantity of materials as a percentage, facilitating easy comparison across different DCs.
 3. **Dynamic Data:** Visualization updates based on the selected year and quantity threshold from drop-down menus.

- **Interactions:**
 1. **Hover Effect:** Displays a tooltip with material type and quantity when hovering over a battery section.
 2. **Year Drop Menu:** Allows users to select a specific year to display different quantities, including historical data and future forecasts.
 3. **Threshold Alerts:** Sections with quantities below the selected threshold flash to draw attention.
 4. **Responsive Legend:** Updates based on the selected year, providing a clear reference for the colors used.
- **Gauge Visualization for Inventory Performance:**
- **Visual Encoding:**
 1. **Gauges:** Each gauge represents different inventory metrics (Gross Inventory Quantity, On-Shelf Inventory Quantity, In-Transit Quantity, and Order Quantity), with color-coded needles and arcs.
 2. **Data Utilized:** Represents sales quantities and various inventory metrics, helping compare expected sales with actual performance.
 3. **Needles and Arcs:** Color-coded to represent different metrics, making it easy to interpret performance levels at a glance.
- **Interactions:**
 1. **Dynamic Gauges:** Update based on the selected year, showing performance levels for different inventory metrics.
 2. **Needles and Arcs:** Represent different inventory metrics, providing a visual representation of performance.
 3. **Legend:** Helps interpret the colors used in the gauges, aiding quick understanding of the data.



Data Used for the Visualization:

- **Inventory Dataset:**
 1. **Gross Inventory Quantity:** Total quantity of a particular item in inventory.
 2. **On-Shelf Inventory Quantity:** Quantity physically present and available for sale on shelves.
 3. **In-Transit Quantity:** Quantity of inventory in transit between locations.
- **Forecast Dataset:** Expected sales for upcoming months.
- **Sales Dataset:** Order Quantity from 5 Distribution Centers for two products: EV Car Battery and Home Battery.

Process Followed to Create the Visualization:

The visualization was crafted through a series of essential steps. Initially, a chart container was established to organize the visual components. Legends were carefully placed within this container to ensure visibility and clear data communication. Dynamic updates were integrated to keep the displayed information accurate and relevant. Furthermore, interactive elements such as hover effects, alert flashing alarms, and dropdown menus were added, enhancing user interaction and simplifying data exploration.

Challenges Faced and Overcome:

During the development of the visualization, we faced and overcame significant challenges. Integrating historical and forecast data seamlessly was crucial for accurate continuity. We also implemented and tested dynamic features rigorously to ensure smooth interactivity and effective operation under various user interactions.

Specific Features or Techniques Used:

Alert System:

- 1- The alert system uses a color-coded scheme to indicate inventory status: green signifies that both inventory and forecasted quantities are above the threshold, suggesting optimal levels. Orange indicates that one of the forecasted quantities falls below the target, signaling a potential issue. Red alerts that both inventory and forecasted quantities are below the threshold, indicating a serious issue that requires immediate attention.

Differences Between Intended and Actual Design

The actual implementation closely aligns with the intended design, incorporating core elements from the sketches. However, further improvements could include advanced interactivity and real-time data updates.

Code Link:



Conclusion

The implemented visualizations effectively transform the design sketches into interactive and dynamic visual tools. These tools enable users to quickly identify bottlenecks and inefficiencies within the supply chain, enhance inventory management practices, and optimize logistics operations. Further improvements, including advanced interactivity, real-time data updates, and customizable parameters, are recommended to enhance the performance of the supply chain process and overall company performance.

Implementation 2: Plant Shipment Performance to Distribution Centers

Description of the Visual

The implementation for visualizing plant shipment performance to distribution centers (DCs) involves two maps: a world map and a map of the European Union (EU):

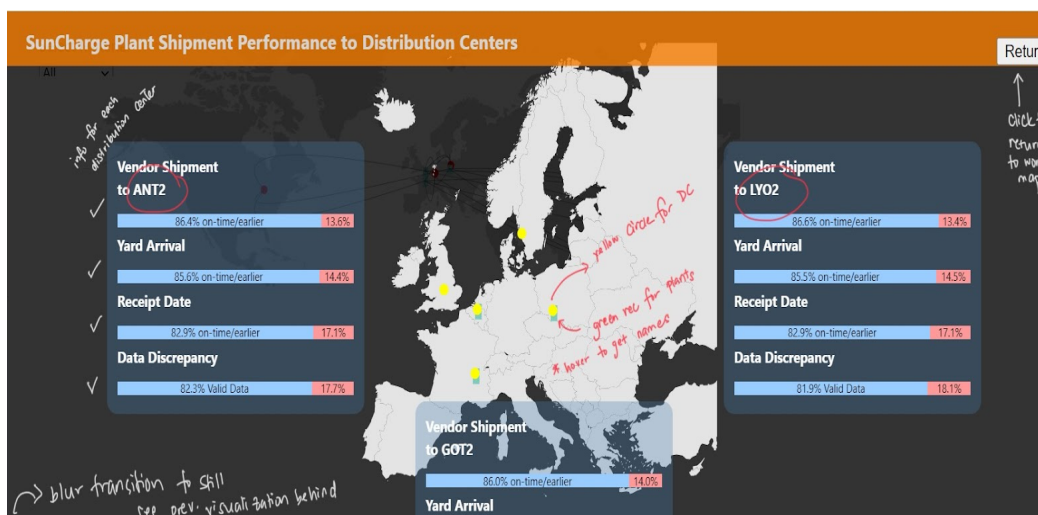
Map1- World Map

This map displays a global view where red circles indicate External Vendors and green rectangles denote the Plants. The splines connecting External Vendors to the Plants illustrate the relationships between them, specifically depicting the Total Inbound Lead Time. This lead time encompasses production time, inbound transportation, and goods receipt processing, providing a comprehensive view of the supply chain logistics.



Map2- EU Map

On the EU Map, the layout features green rectangles representing the Plants/Internal Vendors, while yellow rectangles indicate the Distribution Centers. This map focuses on the logistics infrastructure within the European Union, highlighting the distribution network and plant locations.



Intended Design

- **Sketch Reference:**
 - 1- **Converge 1- Converge 1: Vendor-to-Production Relationship Map and Modified Sankey Diagram:**
 - Snapshot of relationships between each external vendor and the plants.
 - Improved with a material drop-down button to filter and view specific relationships better.

Actual Design

The implemented visualizations closely follow the intended design, incorporating dynamic and interactive features to enhance usability and insight generation.

- **World Map:**
 - **Visual Encoding:**
 1. **Red Circles:** Represent external vendors.
 2. **Green Rectangles:** Represent plants.
 3. **Splines:** Connect external vendors and plants, showing Total Inbound Lead Time.
 - **Interactions:**
 1. **Hover Effect:** Resizes the red circles/external vendors and shows the name and location information.
 2. **Hover Effect on Plants:** Shows names and location of the plants/green rectangles.
 3. **Material Drop-Down Menu:** Filters and displays splines for specific vendors supplying that material.
 4. **Click on Plants:** Transitions to the EU map with the background darkened.

This map features interactive elements such as hovering over the red circles, which represent the External Vendors, to resize them and display the vendor's name and location. Similarly, hovering over the green rectangles, which depict the Plants, shows their names and locations. Additionally, a dropdown menu for selecting Materials is located in the top left corner below the header. Choosing a material from this menu highlights the splines connecting specific vendors that supply the selected material. Clicking on any of the Plants causes the EU Map to appear overlaid on the World Map with a blurred background and reduced opacity, creating a smooth transition between the two maps.

- **EU Map:**
 - **Visual Encoding:**
 1. **Green Rectangles:** Represent plants/internal vendors.
 2. **Yellow:** Represent distribution centers.
 3. **Overlay Boxes:** Contain bars with information on shipping times to each of the 5 DCs.
 - **Interactions:**
 1. **Hover Effect:** Shows names and location for both plants and distribution centers.
 2. **Return Button:** Allows returning to the world map visualization.

In the transition to the EU Map, interactive elements include hovering over both the Plants and Distribution Centers to display their names and locations. Moreover, this map displays five overlay boxes containing bar charts with shipping times to each of the five Distribution Centers, strategically positioned using z-index for clear visibility. A 'Return' button is placed in the top right corner, allowing users to easily switch back to the World Map visualization.

The initial sketch captured a snapshot of the relationships between each external vendor and the plants. In the final implementation, this concept was enhanced by introducing a Material dropdown button, which allows users to filter and better visualize these relationships. However, certain features were not included in the final version. For example, animated splines moving from point A to point B were omitted, as this would require the expertise of a skilled programmer. Additionally, the idea of highlighting splines upon hovering was not implemented due to the complexity of coding multiple paths for each of the 30+ materials. The transition to the EU Map was an effective upgrade from the static sketch, enhancing the interactive experience. Nonetheless, the addition of more filters, such as by year or order sizes, could have further improved the visualization's utility and user engagement.

Data Used for the Visualization

- **MaterialPlantRelation:**
 1. **Inbound Lead Time Data:** Data detailing production time, inbound transportation, and goods receipt processing.
 2. **Vendor and Plant Data:** Location and performance data for external vendors and plants.

Process Followed to Create the Visualization

To create the visualization, the process began with constructing base maps for both the world and the European Union. We then accurately plotted the locations of vendors and plants on these maps. Following this, interactive elements were added, such as hover effects and transitions, to enhance user engagement and provide dynamic content interaction. Finally, we implemented a material filtering system and ensured that the visualization could dynamically update, allowing users to interactively explore different data perspectives based on selected materials.

Challenges Faced and Overcome:

In addressing the challenges encountered, we prioritized ensuring data accuracy by accurately mapping locations and lead times, which was crucial for the integrity of our visualization. Additionally, we focused on enhancing user experience by implementing smooth transitions and interactive features, which not only made the visualization more engaging but also easier to navigate and understand.

Specific Features or Techniques Used

1. **Material Filtering:** Allows users to view specific vendor-material relationships.
2. **Dynamic Transitions:** Smooth transition between world and EU maps for detailed views.
3. **Shipping Time Overlays:** Provides quick insights into shipping performance to DCs.

Differences Between Intended and Actual Design

The actual implementation closely aligns with the intended design, with enhancements such as the material drop-down button for better relationship filtering. However, it lacks moving splines from point A to point B and additional filters like year or order sizes that could further improve the visualization.

Code Link:

[Redacted Code Link]

YouTube Video Link

[Redacted YouTube Video Link]

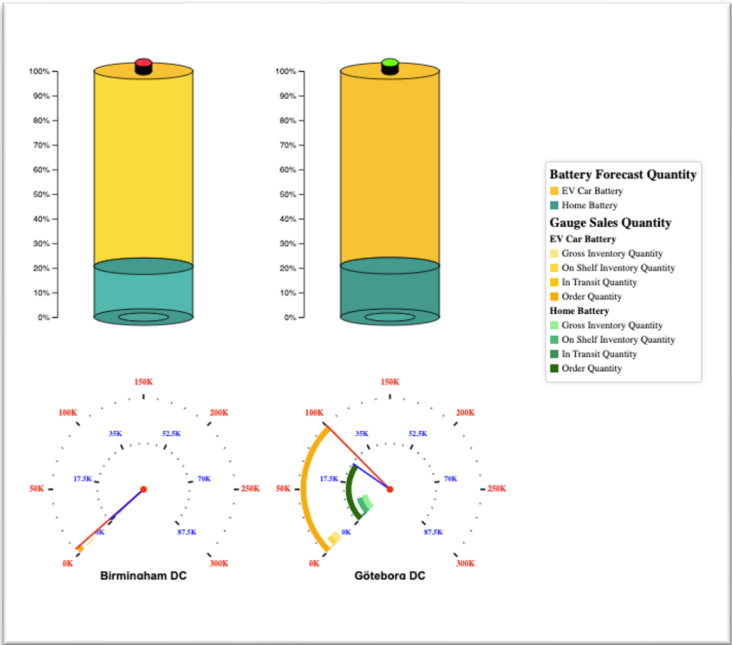
Conclusion

The implemented visualizations effectively transform the design sketches into interactive and dynamic visual tools. These tools enable users to quickly identify bottlenecks and inefficiencies within the supply chain, enhance inventory management practices, and optimize logistics operations. Further improvements, including advanced interactivity, real-time data updates, and customizable parameters, are recommended to enhance the performance of the supply chain process and overall company performance.

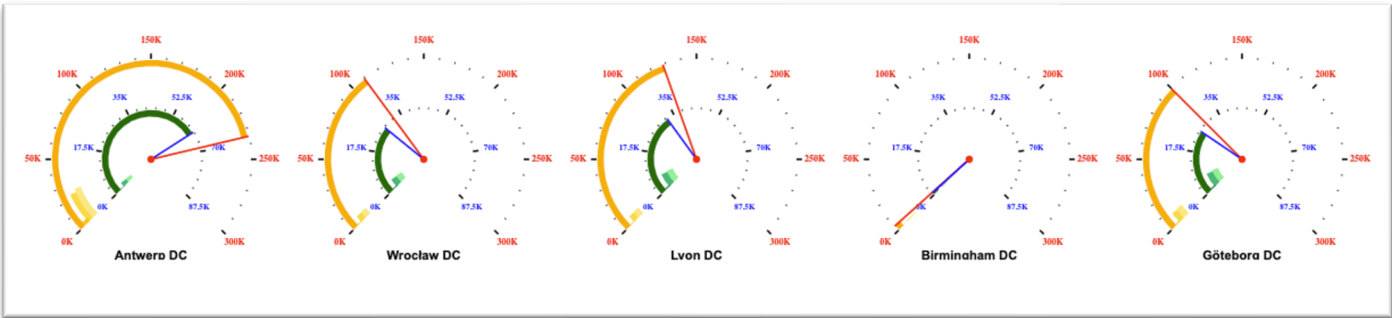
Part 5: Findings

Implimentaion 2: Distribution Centers Performance Monitoring Tool

The performance monitoring tool has revealed that the Birmingham Distribution Center holds significantly fewer products than other centers, a fact highlighted by the red alarm signal indicating lower demand in this region. This observation pinpoints critical areas within the supply chain that require immediate attention. The lower product volume suggests that this center might not be as profitable as others, highlighting a need for increased investment to enhance sales and customer engagement locally. Additionally, discrepancies between forecasted and actual sales quantities call for a thorough verification of the data to ensure its accuracy.



Other trends identified include forecasted data aligning with sales quantities. However, a closer examination of the gauge reveals that the gross quantities available—whether on shelf, ready for shipment, or in transit—are substantially lower than the quantities ordered by sales. This discrepancy signals a need for the supply chain to enhance the production phase to ensure output meets demand or to investigate the causes of inventory shortages, such as potential delays in delivering products to inventory and preparing them for distribution. Furthermore, a thorough double-check of the data's correctness and reliability is essential to maintain accurate supply chain management.



Implimentaion 2:

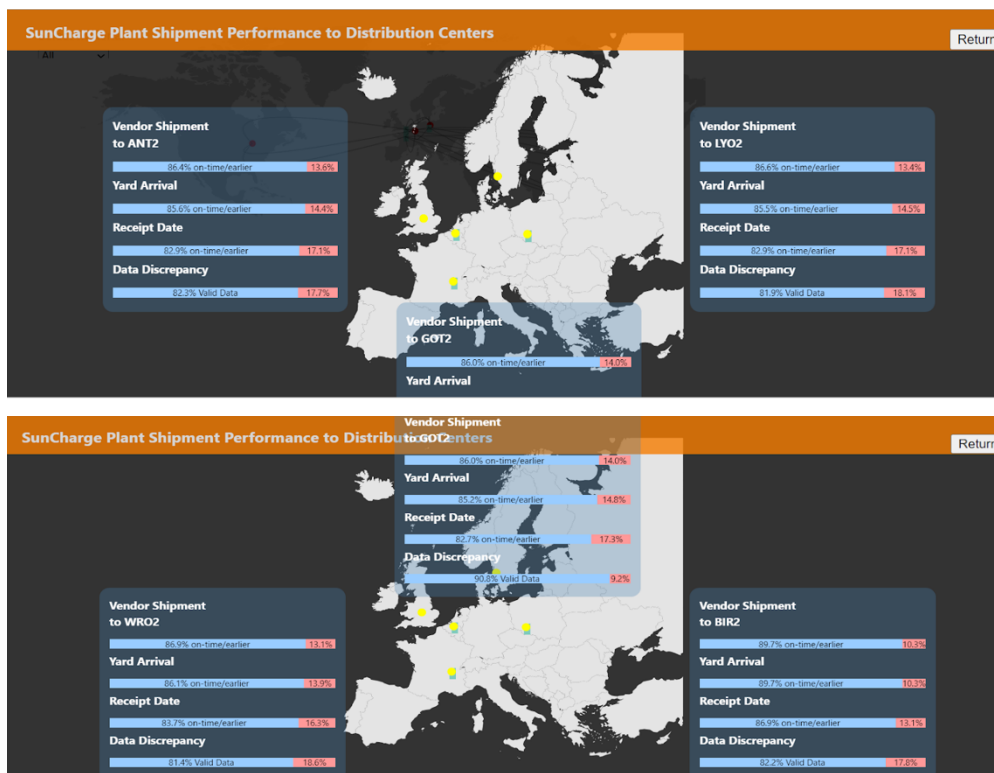
The visualization of external vendors to plants highlights that Materials 3 (Semi-Finished Battery Cells) and 4 (Semi-Finished Battery Management System), supplied by Vendor 5 (DragonCell Energy in Shanghai, China) and Vendor 6 (JadePower Battery in Miaoli, Taiwan) respectively, have the longest Total Inbound Lead Time. Conversely, Materials 24 (Electrical Material 3) and 25 (Electrical Material 4), categorized as Miscellaneous/Other Electrical Components and supplied by Vendor 2 (Stiedenann in Dortmund, Germany) and Vendor 3 (Walker-Homenick in Poznan, Poland), have the shortest lead times, typically around two days.

The data also reveals that the difference in total inbound lead times between plants is generally minimal, often just one day, influenced primarily by the distance from the external vendor.

To enhance efficiency in the supply chain:

- Materials 3 and 4, as critical components with lengthy production and delivery times (approximately 30 days each), should be ordered well in advance. Implementing nearby warehouse storage and maintaining a safety stock for these parts could mitigate unexpected delays.
- Fast-moving items, such as Materials 24 and 25, can be ordered on an as-needed basis to reduce storage costs.
- The dependency on single external vendors for some materials poses a significant risk to business continuity. It would be prudent to diversify suppliers, particularly for essential components, alongside stringent quality control measures.

In the visualization from Plant/Internal Vendor to Distribution Shipping, about 15% of the records display inconsistencies where the Shipping Date is later than the Receipt Date, undermining data reliability for decision-making. Notably, Distribution Center GOT2 exhibits a substantially lower percentage of data errors. Adopting record-keeping best practices from this center could enhance data accuracy across other centers.



The average late shipment rate from plants to business centers stands at approximately 14%, with the vendor for BIR2 showcasing the best performance in timely shipments. Interestingly, both Distribution Centers ANT2 and BIR2 receive supplies from Plant ANT1, yet BIR2 consistently shows better processing times despite its greater distance compared to ANT2. This discrepancy raises questions about internal efficiencies.

Given the challenges in pinpointing shipping delays with the current data, further investigation into production bottlenecks, supply shortages, transportation issues, and alternative delivery options is warranted. Such measures are crucial as they impact the Yard Arrival Date and Goods Receipt Date, potentially affecting the entire supply chain.

Part 6. Reflections

What are we most proud of in this project?

We are most proud of our team's ability to transform initial design sketches into functional and interactive visualizations that effectively address the key questions of our project. The Distribution Centers Performance Monitoring Tool, with its dynamic updates, interactive features, and clear visual encoding, stands out as a significant achievement. It was rewarding to see how our collaborative efforts and iterative feedback loops resulted in a tool that provides valuable insights into supply chain performance. Additionally, our successful implementation of the alert system and comparative analysis features in the Gauge Visualization for Inventory Performance demonstrated our technical skills and attention to detail. This project has showcased our ability to turn complex data into intuitive and actionable visualizations, which is a crucial skill in data visualization and analysis.

What are we least proud of in this project?

While we are proud of our overall achievements, we are least proud of the limitations we encountered due to technical constraints and time restrictions. In particular, some advanced interactive features and real-time data updates that we envisioned in the design phase were not fully realized in the actual implementation. For instance, the moving splines in the World Map visualization and additional filters like year or order sizes in the EU Map were not incorporated. These features would have enhanced the user experience and provided deeper insights. Additionally, some aspects of the visualizations, such as the dynamic transitions and hover effects, could be smoother and more responsive.

Another significant challenge we faced was coordinating our work across different time zones, with an 8-hour difference between team members. This made it difficult to find matching and suitable times for interaction, often delaying feedback and collaboration. Overcoming this hurdle required extra effort in planning and communication, and while we managed to navigate this challenge, it added a layer of complexity to the project. Addressing these areas in future iterations or projects would significantly improve the overall quality and effectiveness of our visualizations.

Part 7. Individual Contributions

Both team members contributed significantly to the sketching, design process, and implementation phases of the project. Having only two team members limited the brainstorming for a wider array of ideas, but this constraint stimulated our creativity to overcome this limitation. Additionally, working across different time zones added an extra challenge, with an 8-hour difference requiring careful planning and communication.

Joana was particularly inspiring with her designs, which helped to accelerate the project's progress and facilitated the improvement, adaptation, and modification of the designs. I, Joana, worked on the first implementation: the Distribution Centers Performance Monitoring Tool. Joana worked on the second implementation: Plant Shipment Performance to Distribution Centers. I provided assistance to Joana with interaction ideas to build engaging visualizations.

Each of us also undertook the coding and write-ups for our respective sketches, incorporating ideas generated during our discussions. Additionally, I, Joana, took on the responsibility of final writing and editing, compiling the report into a cohesive document.

In summary:

Joana:

- Contributed heavily to the design and sketching process.
- Developed the second implementation: Plant Shipment Performance to Distribution Centers.
- Brought innovative design ideas that helped to improve and modify the visualizations.
- Managed the coding and write-ups for her sketches.

Joana:

- Contributed heavily to the design and sketching process.
- Developed the first implementation: Distribution Centers Performance Monitoring Tool.
- Assisted Joana with interaction ideas to enhance the visualizations.
- Managed the coding and write-ups for his sketches.
- Undertook the final writing and editing, compiling the report into a cohesive document.

This division of labor and mutual support allowed us to create effective and interactive visual tools that meet the project's objectives.

4. A single overview picture including all sketches that you created. (Appendix A)

For a better visualization, please refer to the following link on Miro: [Miro Board](#)

